POSSESION OF MOBILES IN EXAM IS UFM PRACTICE.

Name _____ **Enrollment No.**_____

Jaypee Institute of Information Technology, Noida

T1 Examination, 2024
B.Tech, VI Semester

Course Title: Computer Networks and Internet of Things
Course Code: 18B11CS311

Maximum Time: 1 Hr
Maximum Marks: 20

CO1	Defining the basics of networking, components and underlying technologies	Remember (Level 1)
CO2	Illustrate the various key protocols in OSI model and TCP/IP protocol suite and explain various application protocols	Understand (Level 2)
CO3	Identification and description of various components, architectures and protocols of Internet of Things (IOT) and their real life problems	Understand (Level 2)
CO4	Choose the appropriate network layer and data link layer protocols based on the specific requirements and the communication environment	Apply (Level 3)
CO5	Explain various transport protocols and its performance enhancing mechanisms	Analyze (Level 4)
CO6	Determine the shortest path for the network using various routing protocols and evaluate it	Evaluate (Level 5)

Note: Attempt all the questions.

Q.1. Answer the following question briefly: **[CO1, (Remember), 4 Marks]**

a) When a host on network A sends a message to a host on network D, which address stays the same between the routers?
b) Which layer ensures error-free data unit delivery from one station to the next?
c) Which layer of B reads the header from A's layer 5, when data is transferred from device A to device B in the OSI Model?
d) Which protocols control information is transmitted "out of band"?

Q.2. Answer the following question briefly: **[CO5, (Analyze), 4 Marks]**

a) What is the length of the header and the reserved bits for the following Hexadecimal format TCP header 06410028000000010000000600207FF00000000
b) Suppose a user has two browser applications active at the same time, and suppose that two applications are accessing the same server to retrieve HTTP documents at the same time. How does the server tell the difference between two applications?

Q.3. Consider the scenario where the host "jiit.ac.in" seeks the IP Address of the webserver "ox.ac.uk", which is also identified as www.ox.ac.uk. The responses from the Root DNS server follow an iterative approach, while the Top-Level Domain (TLD) response through a recursive method. Following figure shows then DNS hierarchy for "ox.ac.uk". **[CO2, (Understand), 4 Marks]**

a) Provide a sequence detailing the query-response essential for the complete resolution of the requested domain name.
b) What all resource records are placed at TLD DNS server and Authoritative DNS server.

Root

↓

TLD name server (dns.uk, 216.239.31.10)

↓

Authoritative DNS server (dns.ac.uk, 216.239.34.10)

↓

ox.ac.uk (142.250.194.11)

Q.4. Consider the network in figure below. A file of size $1*10^5$ bits needs to be sent from node S to node D. Each link has bandwidth of 1 Kbps and the propagation delay for each link is 0.002 sec. Suppose packet header is of 20 bits and packet size is 1024 bits. Ignore processing and queueing delay. [Hint: Packet size= Header + Payload]

a) If data file is being sent from S to D using packet switching, calculate end-to-end delay for packet (datagram). **[CO1, (Remember), 3 Marks]**

b) Calculate the total delay for the file to arrive at D, if the user file is sent using virtual-circuit switching and call set up time is 0.5 sec. **[CO1, (Remember), 1 Marks]**

Q.5. Consider the scenario, where an HTTP server and client interact. Assume a round-trip time (RTT) delay of 20 ms between the client and the server, a transmission time of 0.25 ms for the server to send an object through its outgoing link, and negligible transmission time for any other HTTP message that does not contain an object. Suppose the client initiates 100 consecutive requests, waiting for a response before sending the next request. The client utilizes the IF-MODIFIED-SINCE Header line. It is also assumed that 60% of the requested objects remain unchanged since the client initially downloaded them (prior to the execution of these 100 downloads). Calculate the total time required to transmit the object using: **[CO2, (Understand), 4 Marks]**

a) HTTP 1.0
b) HTTP 1.1